

12.151

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Question

A point **P**(10, 3) MPa on the Mohr's circle represents normal and shear stresses. If the centre of the Mohr's circle is **C**(6, 0) MPa, the normal and shear stresses in MPa on the point diametrically opposite to **P** are

- ① 2, -3
- ② 4, -3
- ③ 2, 3
- ④ 4, 3

Theoretical Solution

A point on the Mohr's circle,

$$\mathbf{P} = \begin{pmatrix} \sigma_P \\ \tau_P \end{pmatrix} = \begin{pmatrix} 10 \\ 3 \end{pmatrix} \text{ MPa} \quad (1)$$

where,

σ = Normal Stress and τ = Shear Stress

The center of the Mohr's circle,

$$\mathbf{C} = \begin{pmatrix} \sigma_C \\ \tau_C \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \text{ MPa} \quad (2)$$

Let P' be the point diametrically opposite to P ,

$$\mathbf{P}' = \begin{pmatrix} \sigma'_P \\ \tau'_P \end{pmatrix} \text{ MPa} \quad (3)$$

Theoretical Solution

The center **C** is the midpoint of the line segment connecting **P** and **P'**. We can use the midpoint formula for both the horizontal (normal stress) and vertical (shear stress) coordinates.

$$\mathbf{C} = \frac{\mathbf{P} + \mathbf{P}'}{2} \quad (4)$$

$$\mathbf{P}' = 2\mathbf{C} - \mathbf{P} \quad (5)$$

$$\mathbf{P}' = 2 \begin{pmatrix} 6 \\ 0 \end{pmatrix} - \begin{pmatrix} 10 \\ 3 \end{pmatrix} \quad (6)$$

$$\Rightarrow \mathbf{P}' = \begin{pmatrix} \sigma'_P \\ \tau'_P \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \text{ MPa} \quad (7)$$

